

Background

How variable are the object categories that children actually see?

- Objects in the child's view follow a skewed frequency distribution; frequent categories are learned earlier (Clerkin et al., 2017; Clerkin & Smith, 2022; Yang et al., 2026)

- Learning from more variable exemplars supports generalization to new category members (Perry et al., 2010; Twomey et al., 2018)

- But equally frequent categories can differ in visual variability, this variability has rarely been characterized in children's naturalistic input

BabyView Dataset

- A large, longitudinal dataset of everyday experience from the young children's perspective

- Vertical FoV captures object interactions

- 868 hours of data (release 2025.1).

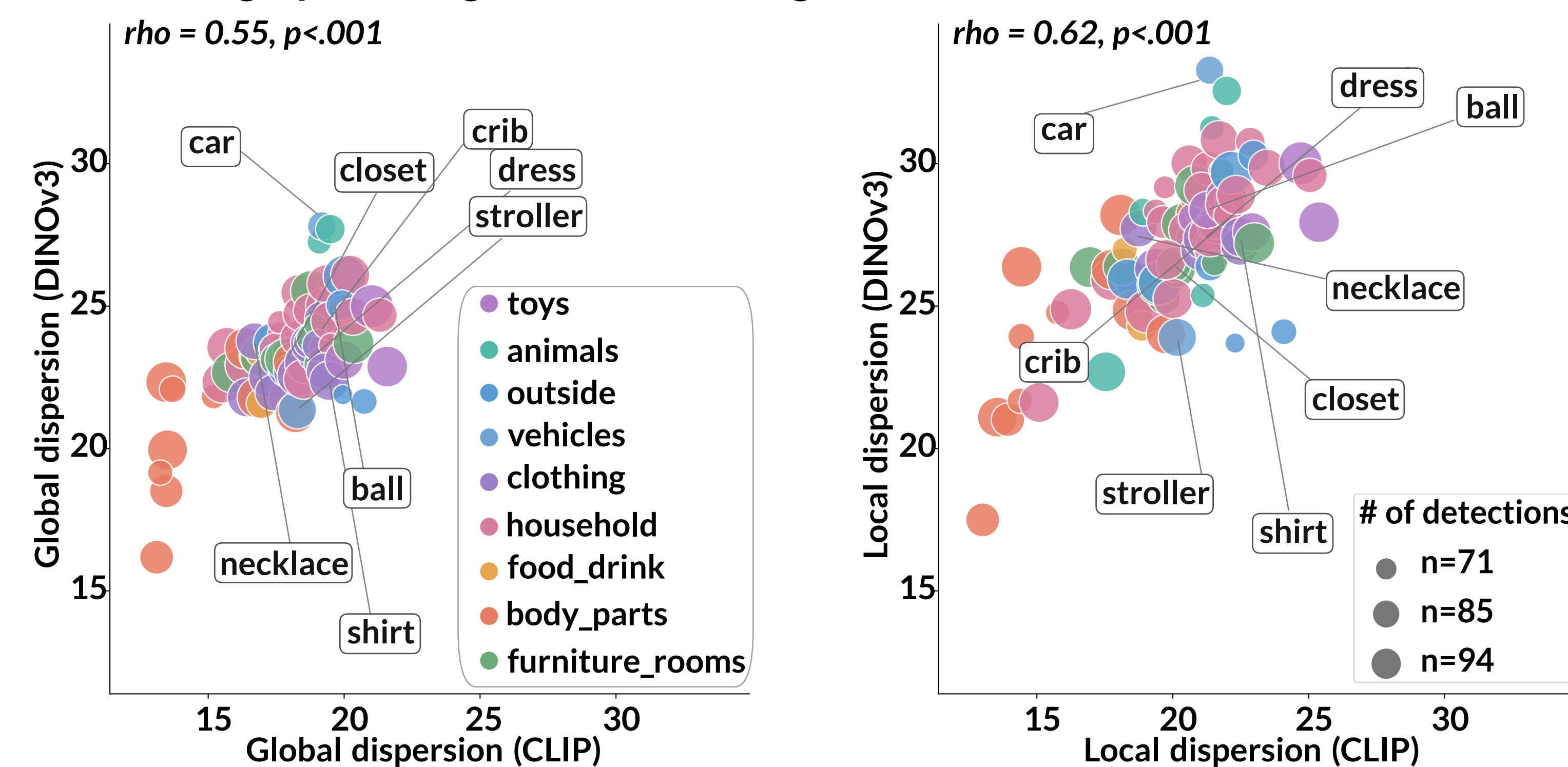
- $N = 31$ (5-36 months) (Long et al., 2025; Yang et al., 2026)

Results

Object categories vary widely in everyday visual variability



Models largely converge on which categories are most variable



Conclusions

- Everyday categories differ widely in within-category variability

- Ranking partially replicates across CLIP and DINOv3

- Variability is weakly, inconsistently coupled to frequency

- Local and global dispersion diverge for some categories, and relate differently to repeated views

Future directions

- How does everyday visual variability shape the trajectory of early concept learning?

- Which dimensions of variability matter most: viewpoint, instance diversity, or context?

- Does everyday visual variability change across development, and does it track how categories are learned?

interactive data page

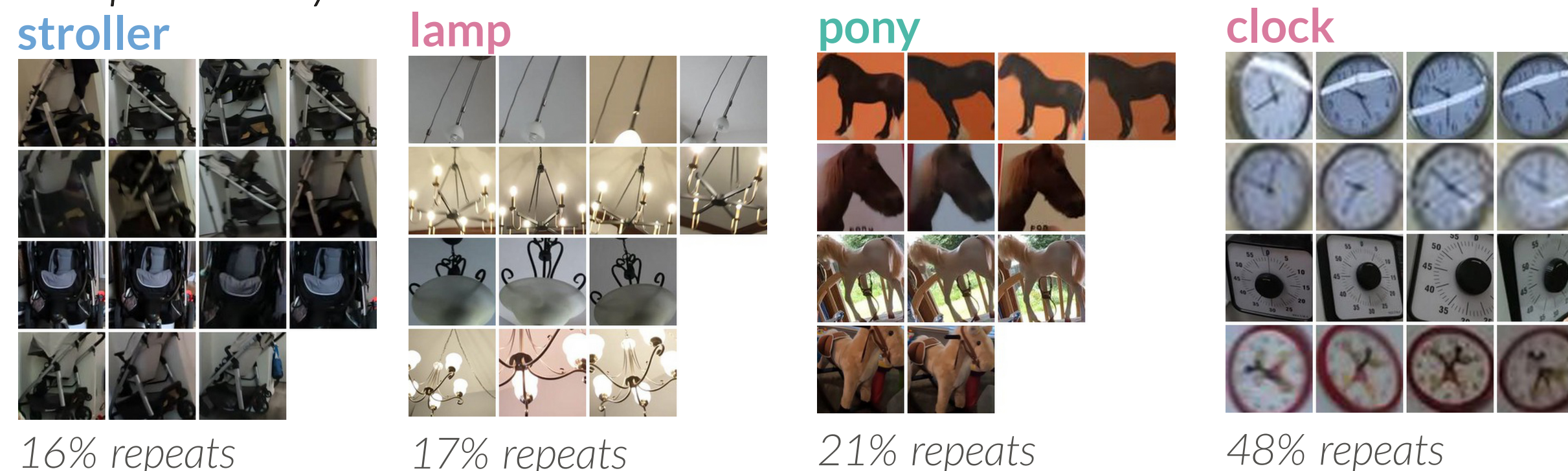
abstract & code repo

objects preprint

Ongoing analyses

Categories range from viewpoint-driven to instance-driven variability

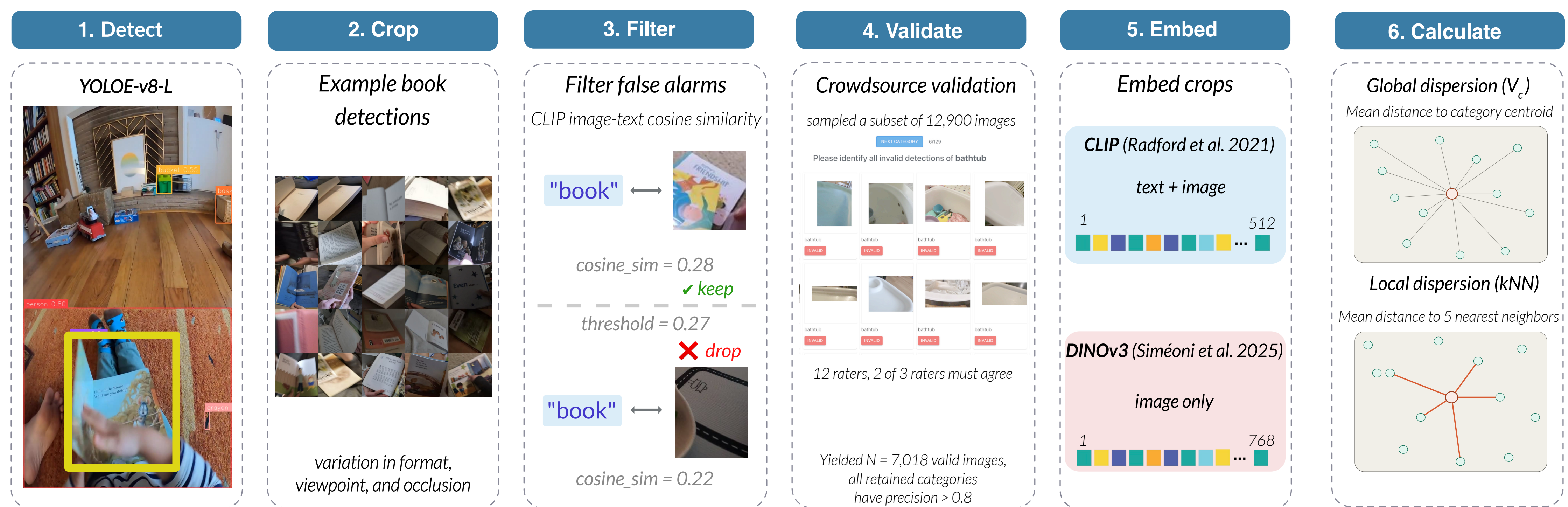
Viewpoint-heavy



Instance-heavy



We detect objects in the child's view, validate the detections, and embed each exemplar to measure how visually variable each category is



Model choices

- CLIP: captures semantic and format structure (Radford et al., 2021)

- DINOv3: visual appearance alone (Siméoni et al., 2025)

Key measures

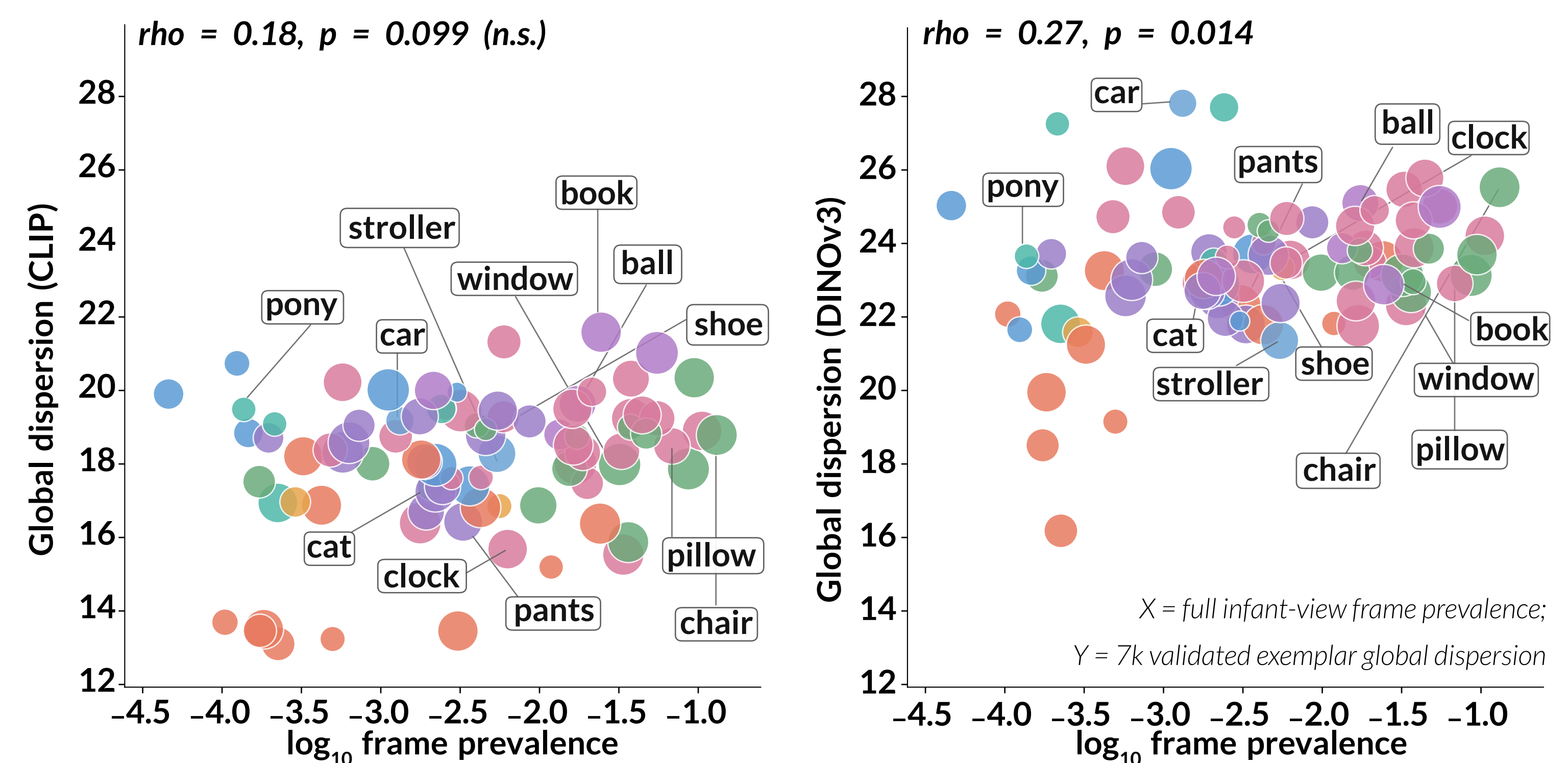
- Global dispersion (V_c): higher = more spread overall (i.e., bigger category cluster)

- Local dispersion (kNN): higher = neighbors more spread out (i.e., less repeats)

All analyses were done on 85 CDI noun categories, $N = 7,018$ human-validated crops (57–100 per category)

Ranked from low to high using overall variability (Dinov3)

Frequency only weakly predicts how variable a category looks



All preliminary analyses were done on 72 non-body-parts categories for privacy concerns, $N = 5,912$

Local dispersion tracks repeated views; global dispersion appears unrelated

